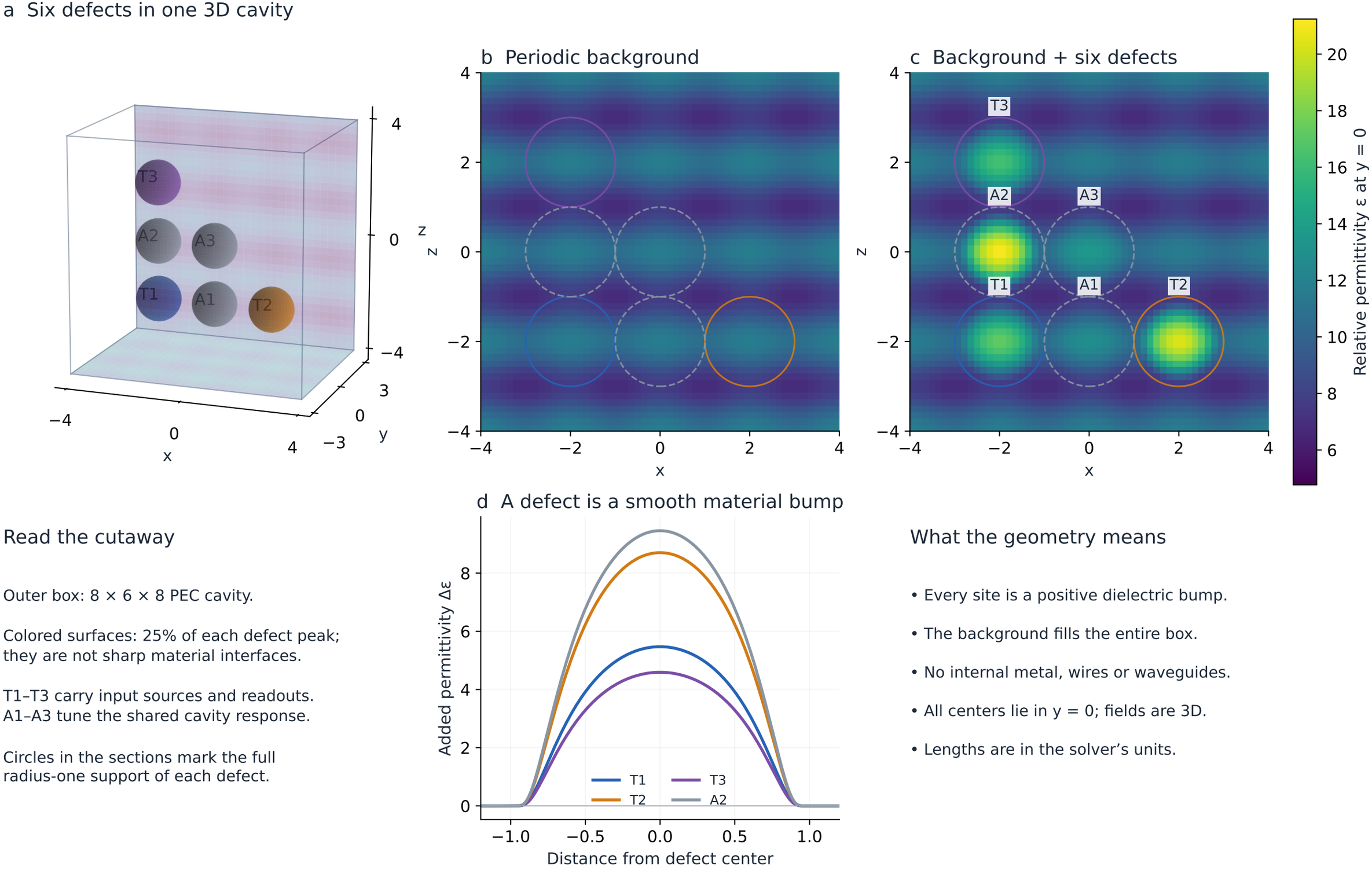

a Six defects in one 3D cavity
T3
A2
A3
T1
A1
T2
x
y
z
b Periodic background
c Background + six defects
Relative permittivity ε at y = 0
d A defect is a smooth material bump
Added permittivity Δε
Distance from defect center
T1
T2
T3
A2
Read the cutaway
Outer box: 8 × 6 × 8 PEC cavity.
Colored surfaces: 25% of each defect peak; they are not sharp material interfaces.
T1–T3 carry input sources and readouts.
A1–A3 tune the shared cavity response.
Circles in the sections mark the full radius-one support of each defect.
What the geometry means
• Every site is a positive dielectric bump.
• The background fills the entire box.
• No internal metal, wires or waveguides.
• All centers lie in y = 0; fields are 3D.
• Lengths are in the solver's units.